

# Logan Lane

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## EDUCATION

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### Virginia Tech

Ph.D. in Computer Science and Applications advised by Doug A. Bowman  
Thesis: *Improving 3D Interaction Through Models of Task Completion Time*

Blacksburg, VA, USA

Aug. 2021–Aug. 2026 (Expected)

### Virginia Tech

M.S. in Computer Science and Applications - *Earned en route to Ph.D.*

Blacksburg, VA, USA

Aug. 2021–May 2024

### University of Virginia's College at Wise

B.S. in Management and Information Systems  
Minors in Computer Science & Business Administration

Wise, VA, USA

May. 2018–May 2021

## EXPERIENCE

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### Virginia Tech

Graduate Research Assistant

Blacksburg, VA, USA

May. 2022–Current

- Conducted research that developed and validated performance models to predict user interaction time for 3D User Interfaces through a grant from the Office of Naval Research
- Conducted additional VR/AR research spanning a range of topics including collaboration, training, and general 3D interaction techniques
- Developed prototype VR/AR software using C#, Unity, Unreal Engine, and Photon/Normcore Networking technologies for Meta Quest and Magic Leap Head Worn Displays
- Investigator on multiple HCI studies; recruited and ran nearly 200 participants

### Prosimulation, LLC.

Founder/Developer

Blacksburg, VA, USA

March. 2023–October 2025

- Served as PI on a \$50,000 proof-of-concept grant and directed a five-developer team
- Achieved 90.1 mAP@50 detecting football players by fine-tuning RF-DETR on 2,000 annotated frames; 84.4% precision and 86.0% recall
- Built a VR analysis tool that ingests RF-DETR outputs to reconstruct and visualize player trajectories in 3D, synchronized to game film for film review and training in an immersive, 3D environment

### NASA

Human Factors Engineer Intern

Huntsville, AL, USA

June. 2024–Aug. 2024

- Developed VR software to simulate lighting conditions on the south pole of the Moon for astronaut training simulation prior to NASA's Artemis mission launch
- Created interactions for Lunar sample collection, shuttle inspection, and photography in VR training simulation using Unreal Engine 5
- Received certificate of recognition for created interactions and mentoring other interns throughout the summer

## SKILLS

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**Programming Languages:** C#, Python, Swift, SQL, Kotlin

**Development Tools:** Unity 3D, Unreal Engine, Photon PUN/Fusion, Normcore, Git, Xcode, Android Studio

**ML, Data & APIs:** PyTorch, vLLM, FastAPI, REST APIs, DuckDB, Docker, CUDA, SQLite

**Platforms:** Meta Quest, Magic Leap 2, Vision Pro, Linux, macOS

## PROJECTS

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### FootballLLM: Natural Language Interface for NFL Analytics

Built a production-style REST API enabling natural language querying of NFL Big Data Bowl tracking data using a two-stage LLM pipeline for text-to-SQL generation and plain English summarization. Deployed via FastAPI with API key authentication, per-request logging, and Docker containerization. Integrated with Virginia Tech ARC's hosted LLM API (Kimi-K2.5, gpt-oss-120b). [github.com/LoganLane/FootballLLM](https://github.com/LoganLane/FootballLLM)

### GPU-Accelerated 3D UI Layout Optimization Pipeline

Developed a CUDA-parallelized pipeline for exhaustive evaluation of 3D user interface layouts using predictive human performance models. Capable of evaluating billions of candidate layouts, with a Python/FastAPI backend and Unity frontend for visualization. DuckDB serves as the analytical backend for querying evaluation results.

### Collaborative, Virtual Reality After-Action Review System for American Football

Implemented a VR Unreal Engine application that allows multiple people to collaboratively review football film in VR.

See full list of projects on <https://loganlane.me/portfolio>

## PUBLICATIONS

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### Journal Papers

1. **Logan Lane**, Feiyu Lu, Shakiba Davari, Robert J. Teather, and Doug A. Bowman, "Revisiting Performance Models of Distal Pointing in Virtual Reality", *IEEE Transactions on Visualization and Computer Graphics*, 2025

### Conference Proceedings

1. **Logan Lane**, Ibrahim Tahmid, Feiyu Lu, and Doug Bowman, "Evaluating the Viability of Additive Models to Predict Task Completion Time for 3D Interactions in Augmented Reality", *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2026
2. **Logan Lane**, Jerald Thomas, Alexander Giovannelli, Ibrahim Tahmid, and Doug Bowman, "Exploring the Effects of Level of Control in the Initialization of Shared Whiteboarding Sessions in Collaborative Augmented Reality", *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2025

Additional publications at <https://loganlane.me/publications>

## SERVICE & PROFESSIONAL DEVELOPMENT

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**Reviewer:** ISMAR 2024, ISMAR 2025, CHI 2025, IEEE VR 2025, CHI 2026

## TEACHING

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### Graduate Teaching Assistant at Virginia Tech

Mobile App Development, Database Management Systems, Professionalism in Computing 2021–2023

## SCHOLARSHIPS AND AWARDS

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|--------------------------------------------------------|------|
| LAUNCH Proof of Concept Grant - \$50,000               | 2025 |
| RADM Fred Lewis Scholarship at I/ITSEC 2024 - \$10,000 | 2024 |